

Michael Gabriel Levy

PhD. Biophysics University of California Berkeley (2019)

Sc.B. Physics *with honors* Brown University (2010) Email: mglevy@berkeley.edu

Research Group Affiliations

Tatyana Sharpee, CNL, Salk Institute (2024-2025) – Dementia/Transcriptomics/Nonequilibrium
Thomas Albright Group, VCL, Salk Institute (2022-2024) – Psychophysics and Neural Modeling
Massimiliano Di Ventra Group, San Diego (2021-2022) – Unconventional Computing
Ian Holmes Group, Berkeley (2020-2021) – Viral Evolution and Bioinformatics
Michael DeWeese Group, Berkeley (2016-2020) – Neuroscience and Physics
Edgar Knobloch Group, Berkeley (2017-2019) – Dynamical Systems and Physics
George Oster Group, Berkeley (2011-2016) – Mathematical Biology
Harold Lecar Group, Berkeley (2011-2013) – Neuroscience and Physics
Biophysics Rotations, Berkeley (2010-2011) – Rokhsar, Isacoff, Theunissen, Geissler Labs
See-Chen Ying Group, Providence (2009-2010) – Polymer Physics
Peter Weber Group, Providence (2007-2009) – Quantum Chemistry

Scientific Publications

- **Levy MG** [*Geometry, Dynamics, Emergence: Cowire Form, Image Geometry, and Coupled Subcritical Oscillations*](#) Dissertation– University of California (2019)
- Rangamani P, **Levy MG**, Kahn S, Oster GF [*Paradoxical signaling regulates structural plasticity in dendritic spines*](#). 2016. PNAS. 113:36
- **Levy MG** *Monte-Carlo Simulation of DNA in Nanopit Arrays* Undergrad Thesis – Brown University (2010)
- Liang X, **Levy MG**, et al., [*Electron diffraction with bound electrons: The structure sensitivity of Rydberg Fingerprint Spectroscopy*](#). 2010. J. Mol. Struct. 978:250
- **Levy MG**, Stratt RM and Weber PM, [*Finite Element Analysis of Molecular Rydberg States*](#), 2009 COMSOL Conference Proceedings

Talks, Presentations, and Workshops

Cellular Energy, Aging, and Optimal Control – 2025 Aspen Center for Physics (Nanoscale Bio)
Distributed Decision Making in Zebrafish – 2023 ODNI Tech Week
Behavioral control of morphology in Cypraeidae – 2021 SICB
Seashell Pattern and Form – 2018 Champaulimaud Research Symposium Travel Award
Cell Cluster Fragmentation – 2018 AIM Invited Pres. (Nonlocal Eqs in Collective Behavior)
Bringing Research into the Physics Lab Course – 2017 AAPT/PERC July Meeting Poster
Developmental Shape Transition in Cowrie Seashells – 2017 APS March Meeting Poster
Modeling of Tissue Growth and Form – 2017 MBI Workshop
Possibilities in Color Blind Camouflage – 2014 UC Berkeley Biophysics Retreat
Neural Mediated Pattern Formation – 2013 UC Berkeley Biophysics Seminar
Neural and Biochemical Circuits – 2012 UC Berkeley Biophysics Seminar
Thesis Talk: Monte-Carlo Simulation of DNA in Nanopit Arrays – 2010 Brown University
Finite Element Analysis of Molecular Rydberg States – 2009 COMSOL Conference Boston

Awards and Honors

Denis Bender Dementia Diagnosis Fund Award (2024) · Intelligence Community

Postdoctoral Fellow (2021-2023) · 2021 SICB Meeting “Travel” Award (2021) · Champaulimaud Research Symp. Travel Award (2018) · Departmental Honors (2010) · Junior Award in Chemical Physics (2009) · George I. Alden Trust Fellow (2008) · Richard Casement Science Writing Fellow – The Economist (2007) · National AP Scholar (2006) · American Legion School Medal Recipient (2006)

Leadership Positions and Departmental Service

- Co-Organizer of the Salk Theory Seminar (2023-25)
- Organizer of the Institute for Neural Computation Chalk Talk Series (2022-25)
- Co-founder of the UC Berkeley Cellular Basis of Patterns Group (2015 – 2019)
- Executive Committee for the Compass Project (2014 – 2016)
- Elected to Executive Committee for Berkeley Biophysics (2011 – 2014)

College Level Course Instruction

- **Head Projects TA for ECOG (NeuroMatch Academy) Summer 2021:**
Prepared open access ECOG data to facilitate group projects across time zones. Also assisted students across the world in developing a research questions and modeling.
- **Co-Instructor Neuro 299: Art of Modeling (UC Berkeley) Spring 2021:**
A practical hands-on introduction to modeling taught native in students’ own research program
- **GSI CogSci 190: Computational Models of Cognition (UC Berkeley) Spring 2018, Spring 2019:**
Historical introduction to AI culminating in Neural Networks and Bayesian Generalization
- **GSI BIOE 231/131: Intro to Computational Mol. And Cell Bio. (UC Berkeley) Fall 2018:**
Practical methods and techniques of bioinformatics for genomics and Metagenomics
- **Head Instructor and Course Design CogSci 190: Data Science and Cognition (UC Berkeley) Fall 2017:**
Designed a hands-on ML research-centric course on categorization and heuristics utilizing NLTK and Network Analysis culminating in independent research using real world datasets.
- **Head Instructor and Course Designer Physics 98: Research Methods (UC Berkeley) Spring 2015:**
Designed a research-centric course on developing scientific questions, conceiving of accessible experiments, and writing results into a scientific paper.
- **GSI for Descriptive Introduction to Physics (UC Berkeley) Spring 2015:**
Ran section teaching out of *Physics for Future Presidents* focusing on scientific literacy
- **Compass Program Summer Teacher (UC Berkeley) Summer 2013 & Summer 2014: & Fall 2015**
Co-Designed and Co-Taught two week courses on *Sound Localization* and *Color, Light, and Materials* for incoming Physics Undergraduates and earned a Mentorship Award for Fall Semester Study Hall
- **Compass Program Teacher Trainer (UC Berkeley) Summer 2014 & Summer 2015:**
Developed and facilitated a 6 week long training for GSI course designers and facilitators
- **GSI BioE 144/244: Introduction to Protein Informatics (UC Berkeley) Fall 2014:**
Applied Machine Learning (mostly Markov Chain Methods) to Evolutionary protein data
- **Intermediate Algebra (Patton University at San Quentin) Fall 2011:**
Co-designed and co-taught Associates Degree level algebra in San Quentin Prison
- **Excursions to Biophysics (Brown University, Providence RI) Fall 2008:**
Designed and taught a computational section in a Physics First Year Seminar

Popular Press Publications

Levy MG and Staff, Origin of Life- [Not that Sinister](#), The Economist (Apr 10th 2008), pp. 89
 Levy MG and Staff, Algal Bioreactors- [Old Clean Coal](#), The Economist Technology Quarterly (Sept 2007), pp. 17
 Levy MG and Staff, Cancer Biology- [Cramping Tumours](#), The Economist (Jan 18th 2007), pp. 73
 Levy MG and Staff, Cosmology - [I spy with my little gravitational lens](#), The Economist (Jan 11th 2007), pp. 92